UNIT 1

PLANTS AND PHOTOSYNTHESIS

Learning objectives:

- Understand the process of photosynthesis and its importance.
- Identify the parts of plant involved in photosynthesis and their roles.
- Recognize the key ingredients for photosynthesis: sunlight, water, and carbon dioxide.

A plant is defined as a living thing that grows on the earth. Plants are an essential part of life on Earth. Plants provide **food, oxygen, fibre, medicine, fuel and shelter for humans and animals.**

Some interesting facts related to plants:

- The plants turn light from the Sun into food required for their growth.
- Plants also need water and nutrients from the soil and carbon dioxide that they take from the air.
- The animals can eat plants so that they can use the food the plants created for their growth too.

DO YOU KNOW?

Why leaves of plants are known as food factory?

Leaves of plants are known as food factories because when they get hungry, they prepare food through photosynthesis, using sunlight, carbon dioxide and water.

One of the most critical processes that plants perform is **photosynthesis,** through which they make their own food.

Photosynthesis occurs in the green parts of plants, mainly in the leaves. Let's learn more about this fascinating process and why it is so important for life?

Basic Parts of Plants:

1. Roots: Absorb water and nutrients from the soil.

2. Stem: Supports the plant and transports water and nutrients.

3. Leaves: Make food for the plant through photosynthesis.

4. Flowers: Produce seeds and fruits.

5. Fruits: Contain seeds and help spread them.

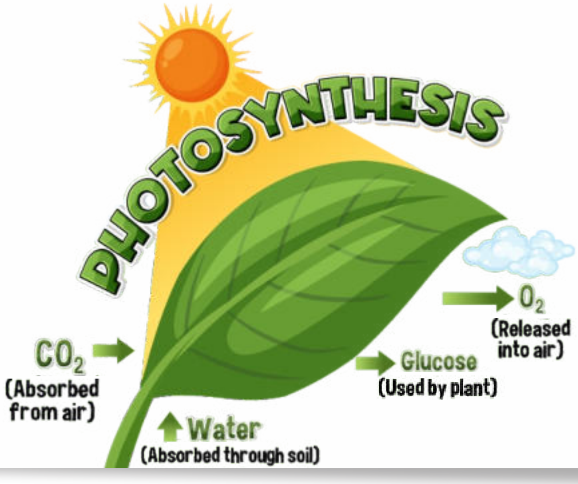
What is Photosynthesis?

The word "**photosynthesis**" comes from two words: "**photo**" (light) and "**synthesis**" (to make).

Photosynthesis is the process by which green parts of plants prepare their food using sunlight, carbon dioxide from the air, and water from the soil. During this process, they release **oxygen**, which is essential for us to breathe.

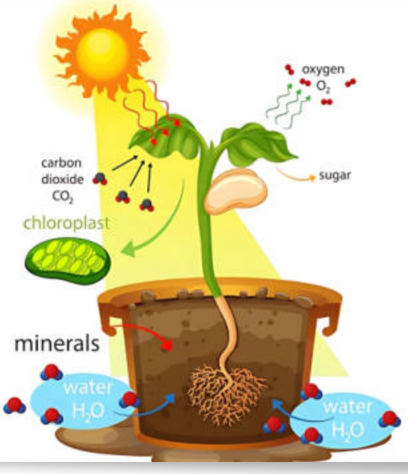
Tiny parts inside leaf cells where photosynthesis takes place is **chloroplast**.

Chlorophyll is a green pigment found inside chloroplast that helps absorb sunlight, which is used to make food for the plant.



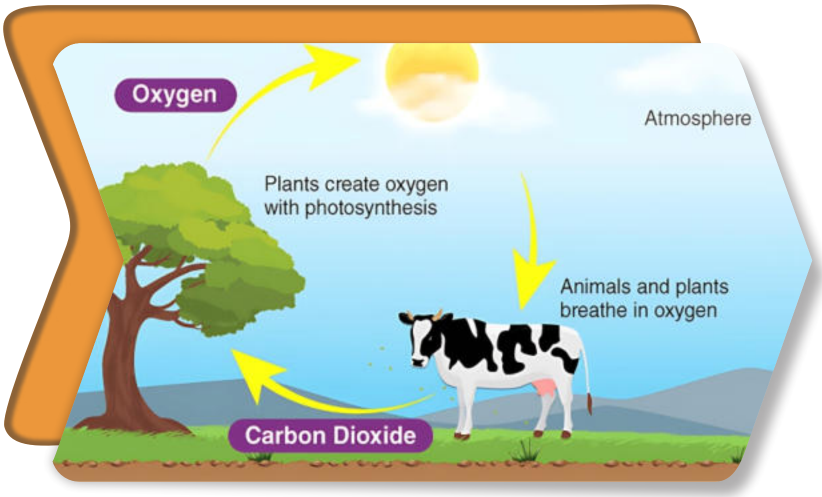
Steps involved in Photosynthesis:

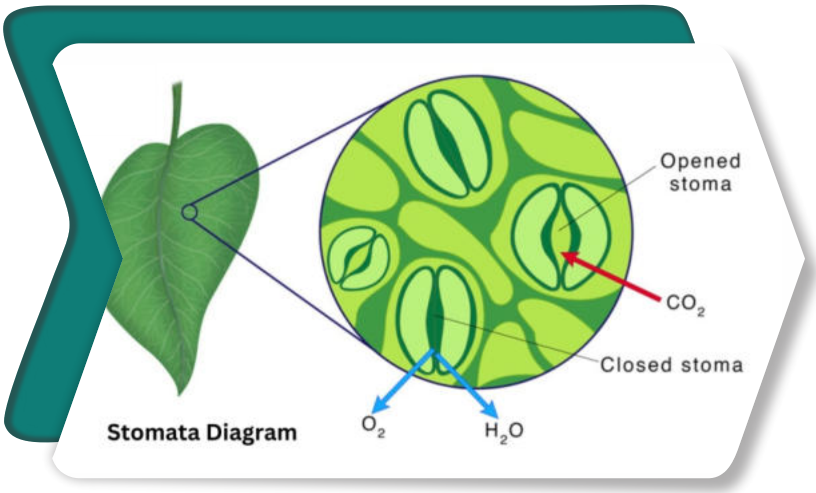
1. Plants absorb water from the soil through their roots.
2. Carbon dioxide enters the leaves through tiny pores called **stomata**.
3. Sunlight provides the energy needed for photosynthesis.
4. Chlorophyll in the leaves uses sunlight to convert carbon dioxide and water into glucose (a type of sugar) and oxygen.
5. The glucose is used as food by the plant, and oxygen is released into the air.
6. Unused glucose is stored in the form of **starch**.



Why is Photosynthesis Important?

1. It produces oxygen, which is essential for life.
2. It forms the base of the food chain, as plants are primary producers.
3. It helps reduce carbon dioxide levels in the atmosphere.
4. It provides the energy needed for the survival of all living organisms.





DID YOU KNOW?

Plants know their season, but they don't use a calendar.

Fill in the blanks.

1. The process by which plants make their own food is called ________________ .

2. Plants absorb water from the ________________ .

3. ________________ is the green pigment in leaves that helps in photosynthesis.

4. Plants release ________________ into the air during photosynthesis.

5. The tiny pores on leaves that allow gases to enter and exit are called ________________ .

Give reason:

1. Why only green parts of plant are involved in making food of plant?

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Write "T" for True or "F" for False.

1. Photosynthesis occurs in the roots of plants.

2. Plants release carbon dioxide during photosynthesis.

3. Chlorophyll absorbs sunlight to help in photosynthesis.

4. Stomata are tiny pores on the leaves.

5. Plants make food at night during photosynthesis.

6. Glucose is the type of sugar plants produce during photosynthesis.

7. Photosynthesis helps reduce oxygen in the air.

8. Carbon dioxide and water are used in photosynthesis.

Choose the correct answer.

1. Which pigment helps plants absorb sunlight?

- Water
- Chlorophyll
- Carbon dioxide
- Stomata

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2. Where does photosynthesis primarily take place?

- Root
- Leaves
- Stem
- Flower

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3. Which gas is released during photosynthesis?

- Oxygen
- Carbon dioxide
- Nitrogen
- Hydrogen

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4. The main source of energy for photosynthesis is?

- Soil
- Sunlight
- Air
- Water

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5. What enters the plant through the stomata?

- Oxygen
- Carbon dioxide
- Glucose
- Starch

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6. The sugar made by plants during photosynthesis is?

- Fructose
- Glucose
- Sucrose
- CO_2

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7. Photosynthesis reduces the amount of ____________ in the air?

- Carbon dioxide
- Oxygen
- Nitrogen
- Hydrogen

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8. Plants store glucose in the form of?

- Protein
- Starch
- Fiber
- Glucose

Answer the following questions.

Q1. Why photosynthesis is important for plants?

Q2. What are the three main things required for photosynthesis to take place?

Q3. Explain the role of leaves in making food?

Q4. Which gas do plants release during photosynthesis, and why is it important for humans?

Q5. What would happen if plants stopped performing photosynthesis?

Critical Thinking

If plants perform photosynthesis to produce food, why is it important for them to have roots, even though they don't make food in the roots?

Create a simple "Photosynthesis Model."

ACTIVITY

- **Materials Needed**: A small potted plant, a clear plastic bag and a rubber band

Steps:

1. Cover a leaf of the plant with the plastic bag and secure it with a rubber band.
2. Place the plant in sunlight for a few hours.
3. Observe the tiny droplets of water forming inside the plastic bag.